DUB-1 CHORD VOICE

ENV again, and again: the stab's sound, its FM, its LFO, its Vary, as of 1.4.0 (build eceec65). FM-1 firmware.



Chord voice 2/4: the LFO on Reso, its reach and its dot on the knob.

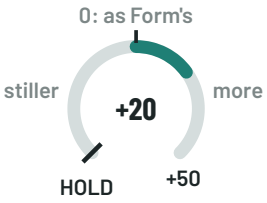


Chord FM 3/4 on Saws: not FM, Colour alone; the others grey.

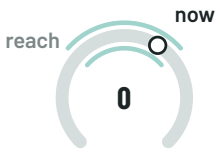


Chord LFO 4/4: what it did, now (the dot), where the wave goes.

VARY: A TRIM



THE MOVING KNOB



its own: its teal's end

1 · THE VOICES ALGORITHM, a chord page

Round, a click a voice; matched in level. Each piece keeps its own. (On the Chord LFO page ALGORITHM is its Target.)

SAWS	Two saws a voice, 7 cents apart.
KEYS FM	An electric piano, brighter as it's struck.
ORGAN	Four drawbars, longer.
PAD	The saws swelling in, long.
BELL FM	FM at 3.5, the clang dying first.
TONEWHEEL	Drawbars, a percussion and a key click at the strike, a rotary speaker turning.
NOISE	Noise and a little saw through the resonant filter.

2 · THE CHORD VOICE PAGE ENV again

Around the voice as it is: your turns stay when the voice changes.

CUTOFF KNOB 1	The low-pass, darker or brighter than the piece's own; each stab opens it further for a moment. `-50 · 0 the piece's · +50 (±2.75 octaves)`
RESO KNOB 2	The filter's resonance, around the voice's. `-50 · 0 · +50`
ATTACK KNOB 3	The strike: 0 the voice's snap, up to a slow swell. `0 – 100`
DECAY KNOB 4	How long each stab rings, times the voice's own (an Organ, a Pad longer). `0 ~60 ms – 100 ~1.5 s`

3 · THE CHORD FM PAGE ENV a third time

Four sine operators (sheet 5 draws the algorithms). On a voice not FM the page says so; Algo, Ratio and Feedback grey and resting.

ALGO KNOB 1	Which operators modulate which: the voice's own, the classic eight, or two. `Voice · 1 – 8 · 2-op`
RATIO KNOB 2	Operators 3 and 4's pitch; grey on two operators. `-50 half · 0 · +50 twice`
FEEDBACK KNOB 3	Op 4 on itself, from the voice's own: pure to gritty. `0 – 100`
COLOUR KNOB 4	Every voice's own: the saws' detune, FM brightness, the organs' upper drawbars, the noise against its saw. `-50 · 0 the voice's · +50`

4 · THE CHORD LFO PAGE ENV a fourth time

Two moves at once: a **wave**, a sine locked to the bars, and a **wander**, slow and random (a new place every 8 to 32 bars, glided to). How far: Form's Drift.

DRIFT KNOB 1	More or less of Form's Drift for this LFO; far left **Still**. The filter about an octave either way at Form's 50 and this 0. `Still · 0 as Drift · +50 twice`
WAVE KNOB 2	The share: wander to wave, the reach the same. `0 the wander · 50 half each · 100 the wave`
RATE KNOB 3	How long the wave, on the bars. A switch. `1 bar (a redonda, whole) · 2 · 4 · 8 · 16 · 32 bars`
TARGET KNOB 4	What it moves (also ALGORITHM). On Reso and Colour as far along their knob (an octave: a 5.5th of the sweep). TO CUTOFF on the rule. `Cutoff · Reso · Colour`
PICTURE	Left, what it did (bold as heard, the wave thin white, the wander pink); the dot now; right, where the wave goes. Its reach at the left edge, in its Target's units (Hz for Cutoff).

5 · THE MOVING KNOB

ON ITS PAGE	The Target's knob (Cutoff or Reso here, Colour on the FM page) shows the LFO: its reach two pale lines either side of the ring, a white dot where it is now.

6 · VARY the Chord page's KNOB 4

A trim on Form's Vary: the chord's amount is Form's plus this. Centred: 0 the notch.

0	As Form's Vary (the pieces' own: 0 to -17).
RIGHT	The stab pattern turns more often (every two bars it may move a step, gain or lose a stab); more throws.
LEFT	Stiller: fewer turns and throws, fewer stabs and bars left out (never more than Form's Vary leaves out).
HOLD	Far left (the word inside the ring): never, whatever Form says. Every stab of the pattern played, no throws but a fill's last.

7 · ITS LEVEL, ITS STRIP OCT · SEL

OCT- / OCT+	On any chord page: its level, 5 a press. Both together: muted, or back.
THE STRIP	SEL, white 1 (F3) held: the chord's Level, Echo, Plate, Filter (sheet 2).
ENV HELD	0.5 s, a **nudge**: Attack, Colour, Reso, Decay, Cutoff (an FM voice: its Ratio, Feedback) a step. 1.5 s, a **shake**: another voice (not Noise), every value a big step.

ENV, AGAIN AND AGAIN

Chord › Chord voice › Chord FM › Chord LFO › Chord; ENV blinks past the first. GLO held 1.5 s: the piece back to its recipe.

THE SCREEN

On the Chord and Chord voice pages, the chord alone: its own ripples in teal (or its scope, its spectrum) and its lane.